

Microplastic Polymer Analysis Report

Report ID	09a6da78-ddd5-4c07-a2ab-8cf67f8422d3
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Sample ID	FTIR-8028

Prediction Results

Detected Polymer	HDPE
Confidence Score	0.94%
Model Used	finetuned_model

FTIR Spectral Analysis

- 2915 cm⁻¹ → CH₂ asymmetric stretching
- 1470 cm⁻¹ → CH₂ bending

Confidence Interpretation

■■■■■■■■■■ 0.94%

Scientific References

[1] Harper, C. Handbook of Plastics Technologies.

[2] Thompson, R. Plastic debris in oceans.

[3] Barnes, D. Accumulation of plastics.

[4] Andrady, A. Environmental impacts of plastics.

[5] Jung, M. FTIR analysis of HDPE.

AI Generated Summary

The analyzed sample was identified as High-Density Polyethylene (HDPE) with a confidence score of 0.94%. The FTIR spectral characteristics strongly support the presence of HDPE polymer chains. The detected CH2 asymmetric stretching at 2915 cm⁻¹ and CH2 bending at 1470 cm⁻¹ are characteristic of the hydrocarbon functional groups in HDPE. This polymer is widely used in plastic bags, bottles, and containers. The analysis indicates a reliable classification suitable for environmental assessment and waste management studies.